Research Project for B.Sc. Degree

An Essay on:

Dye-Sensitized Solar Cell (DSSCs)

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1.Introduction

The urgent global need to reduce greenhouse gas emissions has driven nations to transition from fossil fuels to renewable energy sources. Among these, photovoltaic (PV) technology has garnered significant attention for directly converting solar energy into high-quality electrical energy. PV cells, commonly called solar cells, have evolved through several technological generations Fig.1. **First-generation PV cells** employ single or multi-crystalline silicon and are renowned for their durability and reliability, making them the most widely commercialized **[1]. Second-generation PV cells** utilize thin-film semiconductor materials like cadmium telluride (CdTe) and copper indium gallium diselenide (CuInGaSe2), offering thinner structures but generally lower efficiencies compared to first-generation cells **[2,3]. Third-generation PV cells**, including dye-sensitized solar cells (DSSCs), perovskite solar cells, and organic solar cells, improve flexibility, affordability, and efficiency while employing innovative materials and designs **[4,5].**

DSSCs, a standout technology within third-generation cells, mimic the natural process of photosynthesis and offer unique advantages such as cost-effectiveness, transparency, and flexibility in design. Since their groundbreaking development by O'Regan and Grätzel in 1991**(6)**, DSSCs have demonstrated promising stability and performance, achieving power conversion efficiencies (PCE) of up to 12% for smaller cells. Moreover, their superior performance under low-light and high-temperature conditions makes them a compelling alternative to conventional solar cells. This essay delves into the structural components, operation principles, and materials used in DSSCs, paying special attention to the role of dye sensitizers and their synthesis. The exploration of novel dyes, including natural, metal-free, and

complex metal-based dyes, highlights ongoing advancements to enhance the efficiency , durability, and sustainability of DSSCs.

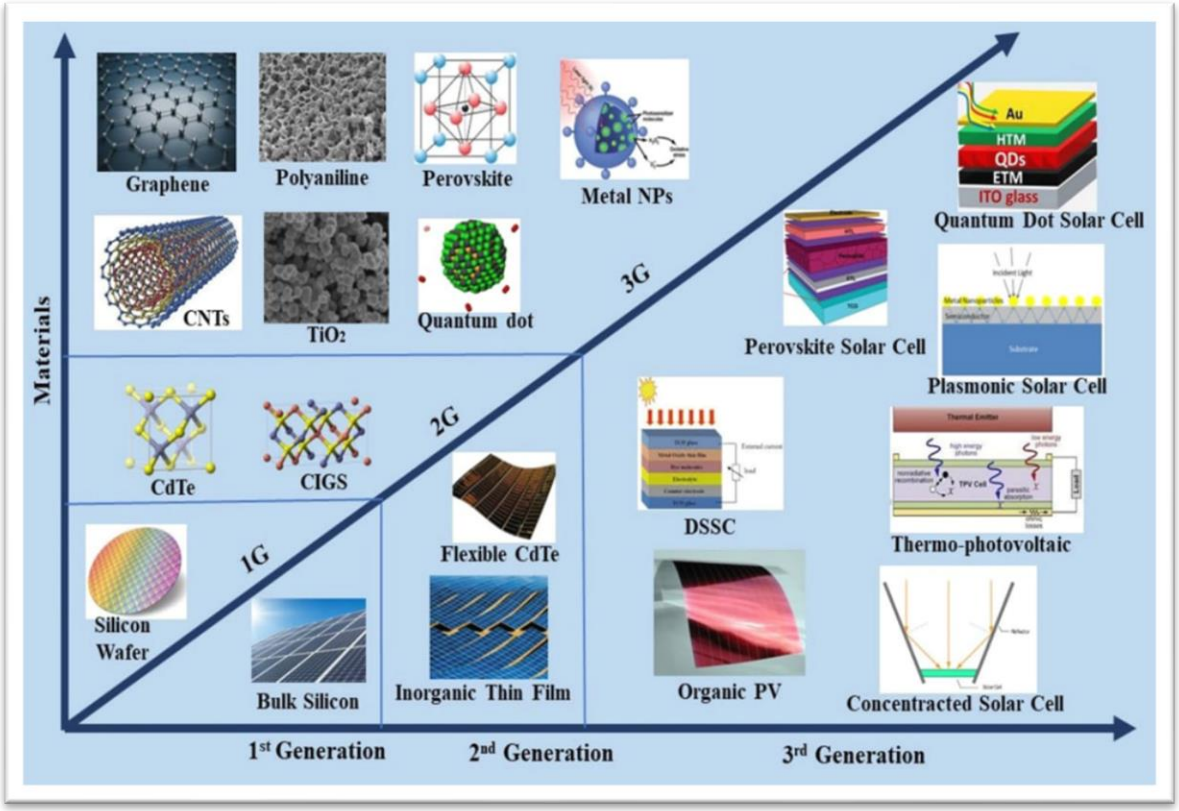


Fig.1 Solar cell development from first, second, and third generation

2. Dye-Sensitized Solar Cells

The new PV cells called DSSCs mimic plant photosynthesis. Unlike traditional solar cells, DSSCs operate better in low light and are less vulnerable to heat-induced energy loss. The efficiency of DSSC is mainly determined by the sensitizer utilized. The dye plays a vital role as a sensitizer in receiving sunlight and converting it into electrical energy **[7]**. DSSCs transform sunlight into electrical power by absorbing photons and producing electrons using dye molecules as shown in **Figure 2**.

To increase the surface area that can be used for adsorbing dyes and so improve the lattice structure and effectiveness of the solar energy harvesting system, it should be

made available in nanostructure form. For successful electron bonding, it must have a conduction band slightly below the dye's LUMO level injection **[8]**. The dye used as a photosensitizer is a vital component of the DSSC process. The absorption spectrum of the dye and its anchoring level to the semiconductor surface have a significant influence on the efficiency of the cell. The chemical structure of dyes is critical for increasing photon absorption and as a result, improves the electron injection and conversion efficiency. Sensitizers, which are the dye molecules, are one of the key elements of the DSSC, encouraging several studies to improve dye efficiency and produce novel dye sensitizer groups **[9]**. To effectively inject electrons from the dyes into the semiconductor's conduction band, an efficient sensitizer for DSSC must

1. bind thoroughly to the semiconductor with an anchoring group, such as a carboxylic or hydroxyl group
2. have a high molar absorbance coefficient due to their deep absorbance in the visible and near-infrared portions of the sun spectrum.
3. transfer electrons from the dye
4. To allow charge injection into the semiconductor, the energy level of the highest occupied molecular orbital (HOMO) must be lower than the redox pairs and the energy level of the lowest unoccupied molecular orbital (LUMO) must be higher than the conduction band level.

Consequently, regeneration of oxidizing dye is feasible. Common sensitizers include metal complex dyes, metal-free organic dyes, and natural dyes **[10]**.

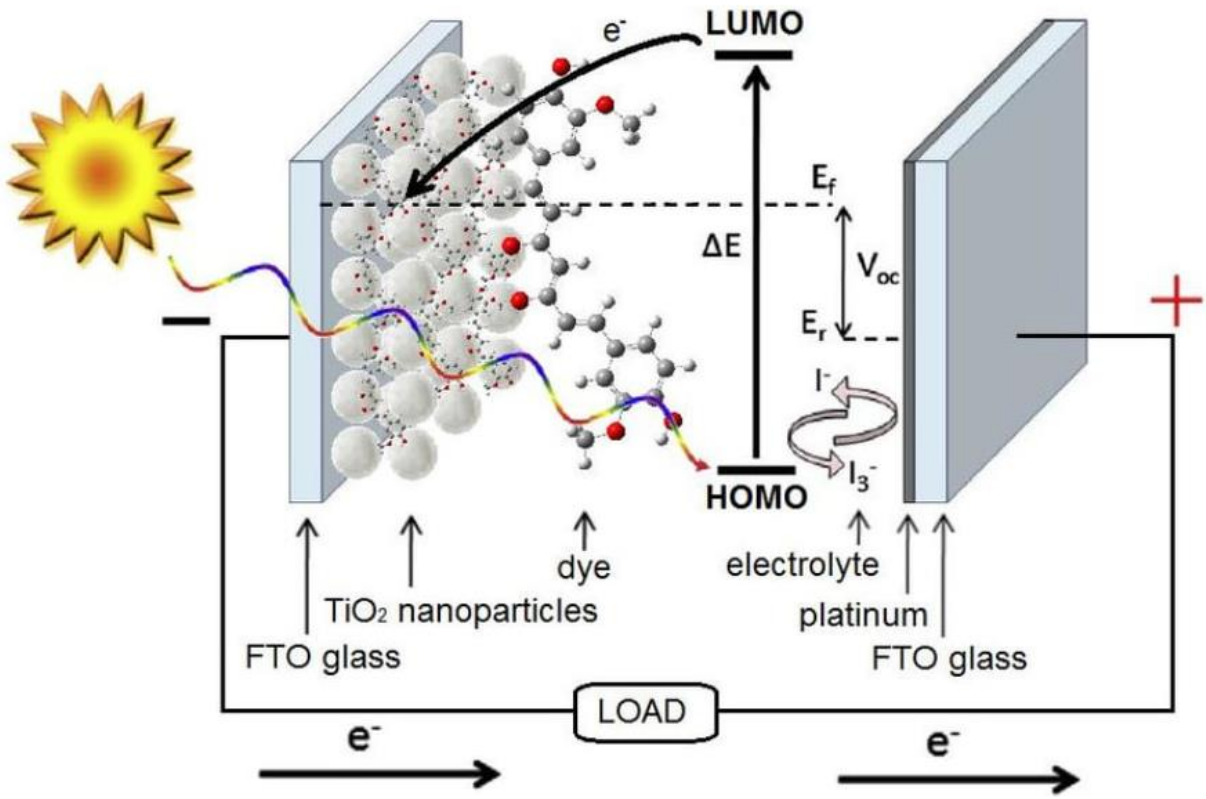


Fig.2 Schematic structure and principle of operation of DSSCs

3. Components of DSSCs

1. Dye Sensitizers

The dye in a DSSC plays a very important role in absorbing solar energy and converting it into electrical energy. By keeping the role of a dye in DSSC in mind, some key conditions are required for a dye to be considered efficient, namely:

1. it must be capable of maximum absorption from the visible region and near-infrared region.

2. it should have excellent binding with anode material

3. the anode material conduction band must be low compared to the LUMO (Lowest Unoccupied Molecular Orbital) of a dye

4. the anode material conduction band must be high compared to the HOMO (Highest Occupied Molecular Orbital) of a dye.

5. it must be durable. **[11]**

Metal complex, natural, and metal-free synthetic organic are the three different types of dye sensitizers. Among the mentioned sensitizers, Metal complex sensitizers exhibit the highest efficiency and greatest stability. However, manufacturing these dyes is difficult, costly, and harmful to the environment. Metal-free synthetic organic sensitizers such as Metal complex sensitizers require a complicated production procedure. On the other hand, natural dyes can easily be extracted from seeds, leaves, fruits, stems, etc. Metal complex and metal-free synthetic organic dye-based DSSCs have high efficiency compared to natural dye-based DSSCs. On the other hand, unlike metal-complex and metal-free synthetic organic dyes, natural dyes do not need any complex preparation process and are cheap, non-toxic, and environmentally friendly **[12].**

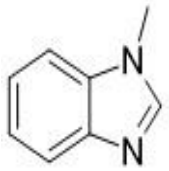
2. Photo-Anode

Semiconductor nanoparticles positioned on a transparent conducting glass that has been dye-sensitized comprise the photo-anode. Many semiconductor oxides, including ZnO, SnO2, WO3, and TiO2, are utilized as materials for photo-anode. TiO2 has drawn the most interest among these semiconductor oxides because of its accessibility, affordability, lack of toxicity, and compatibility.Since a semiconductor serves as a substrate for dye absorption and receives electrons from the dye, its shape is crucial to a DSSC. The dye completes the circuit by introducing the electron into the semiconductor's conduction band upon photoexcitation. From there, it is transferred to the counter electrode. TiO2 nanoparticles are widely used due to their superior crystallinity and dispersion. Three of the most stable polymorphs of TiO2 are anatase, brookite, and rutile; brookite is the least frequent of these due to its intricate synthesis method. Although anatase and rutile perform similarly in terms of solar energy conversion, anatase is preferred over rutile because of its higher electron Fermi level, bigger specific area, and superior charge transport. Furthermore, anatase has a larger bandgap of 3.2 eV as opposed to 3.0 eV for rutile **[13].**

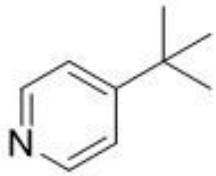
To produce various anatase nanostructures, such as nanorods, nanofibers, and nanowires, researchers are employing a variety of synthesis methods, such as solvothermal, hydrothermal, micelle, inverse micelle, physical vapor deposition, and chemical vapor deposition. An effective electrode contains a scattering layer of about 400 nm and a compact layer of about 20 nm-sized nanoparticles that are 12–14 μm thick **[14].** while the scattering layer is of 400 nm nanoparticles and has a 4–6 μm thickness. Screen printing, doctor blade, spin coating, and spray coating are the different techniques used for coating TiO2 on the conducting glass substrate.

3. Electrolytes

The electrolyte is another crucial element in a DSSC since it can impact a cell's performance and stability. The three main parts of an electrolyte are the redox pair, organic solvent, and additives. Three distinct forms of electrolytes exist: liquid, solid, and quasisolid. Their superior dielectric constant means that. donor count, viscosity, the most utilized solvents in a liquid almost utilized propionitrile, propylene carbonate, acetonitrile, butyrolactone, and Valero nitrile. The redox pair is the electrolyte element that exhibits reducing and oxidizing properties. Because of their improved performance, the redox couples I−/I3− and Co (II)/Co(II) are frequently utilized in DSSCs. Additives are used to stop recombination processes at the contact between the electrolyte interfaces. This is crucial since the Jsc and Voc are decreased by these reactions, which lowers the cell's overall performance. Often used as additives are heterocyclic nitrogen-containing substances like pyridine and its derivatives. The conduction band shifts negatively as a result, increasing the VOC and decreasing JSC. The two additives that are most frequently used are N-methyl benzimidazole and 4-tert-butyl pyridine (tBP). Since they give a positive shift to the conduction band, which raises VOC, compounds like guanidinium and thiocyanate operate in a very different way than the previously discussed additives.



N-methylbenzimidazole



4-tert-butyl pyridine .

4. Counter Electrodes

To rebuild the redox pair, the counter electrode (CE) moves electrons from the outside circuit to the electrolyte. Consequently, the CE needs to be highly stable and have low resistance. A CE material that works well should have a large surface area, strong chemical stability, excellent corrosion stability, high current density, low resistance, and high conductivity.

Because of its exceptional electrocatalytic activity and strong conductivity towards I−/I3−, platinum (Pt) is regarded as an effective CE material. Chemical reduction, chemical vapor deposition, electrochemical deposition, hydrothermal reaction, thermal decomposition, or sputtering are methods that can be used to fabricate Pt CE. Although Pt has great qualities, it is highly expensive and does not work well in many different applications.

As a result, numerous scholars are looking into this. Because of this, a lot of researchers are looking for alternatives that are stable, affordable, and highly efficient. Many carbon types, including carbon nanotubes, carbon nanofibers, and graphite, Pt alloys such as Pt 0.02Co, NiPd, and PtPdNi, alternative metals like Ruthenium (Ru), Iridium (Ir), gold (Au), titanium (Ti), and silver (Ag); polymers like polyaniline and polypyrrole; and bilayers like Au/Pt, Cu/Pt, Ti/Ru, and Al/Ru. When it comes to TiO2 and ZnO, increases the efficiency of the cell. Significantly more effective solar cells were produced using a variety of other approaches, including the molecular engineering of sensitizers, different printing processes, and improvements with anchoring groups.

4. General classes of sensitizers

1. Metal complex dye

Metal complex dyes are frequently utilized as sensitizers because of the high efficiency they attain in DSSC applications.

The metal complex dyes with the highest effectiveness values in DSSCs are (ruthenium bipyridine complex sensitizers N719, N3, Black dye, and Z-907) **[15].**

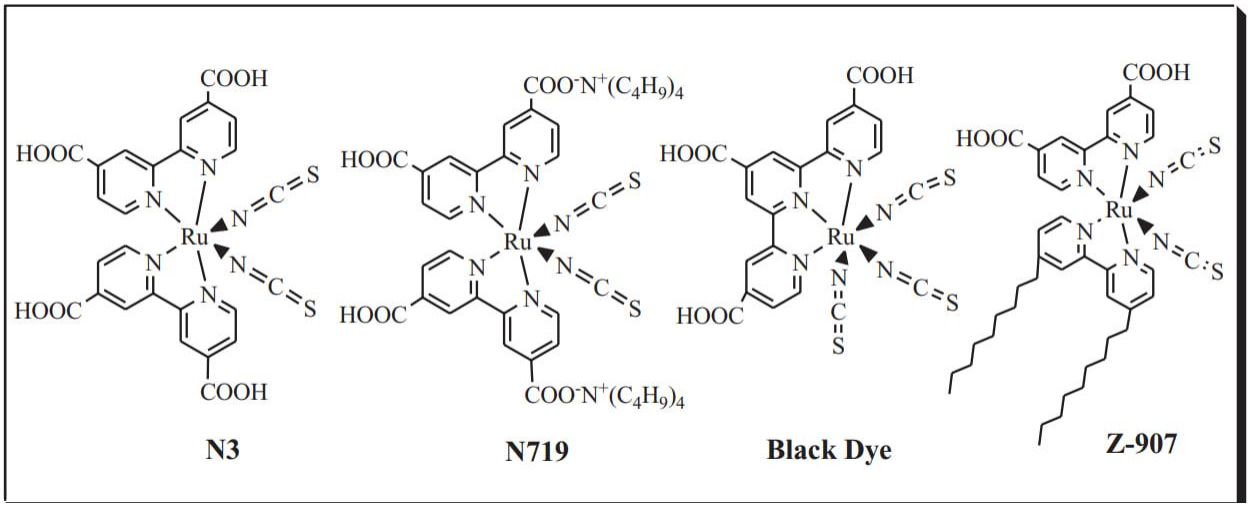


Fig.3 Structures of metal complex dyes used in DSSCs

The Grätzel group's N3/N719 Ru-polypyridyl-complexes have outstanding photoelectrochemical qualities, converting about 11% of the solar energy into electrical energy in the sun's typical global air mass AM 1.5. Ru-based dyes can introduce charges into semiconductors through the phenomenon of metal-ligand transfer. Nonetheless, the Ru complexes contain environmentally toxic heavy metals, and Ru complexes degrade in the presence of water **[16]**. Due to their outstanding stability, recent advancements in ruthenium-based dye compounds have focused on molecular engineering. Ru (II) metal is commonly selected for several reasons:

(1) its octahedral geometrical structure allows for the controlled extension of a specific ligand

(2) Ru (II) complexes' photophysical, photochemical, and electrochemical properties can be modified

(3) it has stable and accessible oxidation states from I to IV

(4) it has good solubility in a wide range of solvents **[17].**

The Role of Metal Complex Dyes in DSSC Functionality

The performance of DSSCs is intrinsically linked to the dye's chemical and photochemical properties. Metal complex dyes, particularly those based on ruthenium, are prevalent because of their high molar extinction coefficients and strong metal-to-ligand charge transfer (MLCT) capabilities. This MLCT enables efficient absorption of light and facilitates swift electron transfer to the semiconductor, ensuring minimal energy loss. Among these dyes, the three leaders, N719, N3, and black dye, serve as benchmarks for advancement in the DSSC realm **[18].**

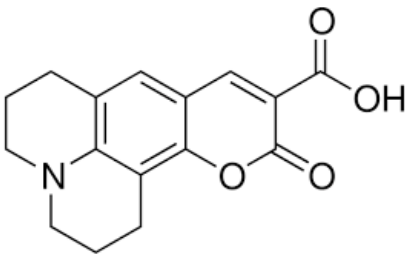
2.Metal-free organic dye

Metal-free dyes are an attractive option for dye-sensitized solar cells (DSSCs) because they avoid the use of heavy metals, minimizing environmental and health hazards while ensuring cost-effectiveness. These dyes are typically organic molecules and often consist of donor (D), π-conjugated (π), and acceptor (A) moieties.

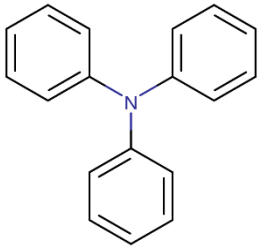
2.1. Organic Dyes

Organic dyes used in DSSCs are based on chromophores such as coumarins, indoline derivatives, triphenylamines, and others, which offer a broad light absorption spectrum:

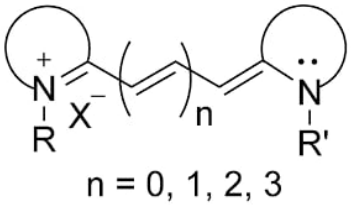
- **Coumarin Dyes**: Known for good π-electron conjugation and electron-donating capability. Examples include C343 and other derivatives **[19].**



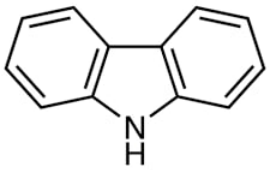
- **Triphenylamine-Based Dyes**: These dyes provide high molar extinction coefficients and are highly tunable for spectral absorption **[20].**



- **Cyanine Dyes**: Used for absorbing in the near-infrared region and enhancing light-harvesting efficiency **[21]**.



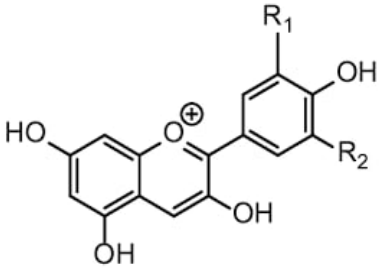
- **Carbazole-Based Dyes**: Excellent photostability and electron-donating properties make them popular for achieving efficient charge separation **[22].**



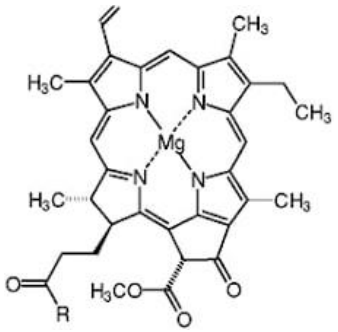
2.2. Natural Dyes

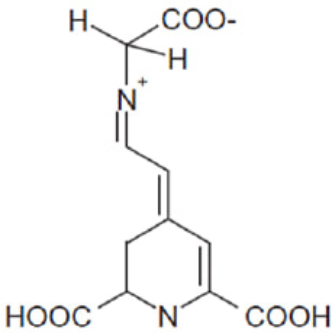
Natural organic compounds extracted from plants, such as anthocyanins, betalains, chlorophyll, and carotenoids, can serve as dyes in DSSCs. Examples include:

- **Anthocyanins:** Found in berries, grapes, and blue flowers, providing deep red to purple hues.



- **Chlorophyll**: Extracted from plants like spinach, it absorbs in the red and blue light regions.





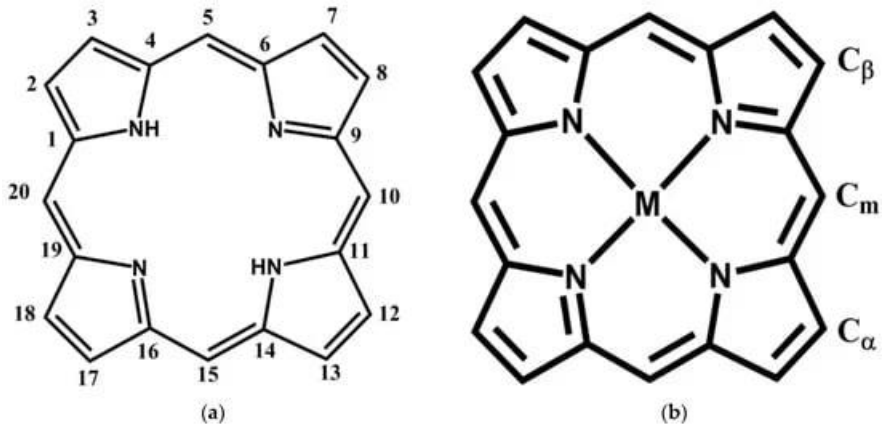
- **Betalains**: Derived from beetroots, these provide high stability and adequate spectral absorption **[23]**.

2.3. Donor–π–Acceptor (D-π-A) Dyes

These dyes utilize a donor (electron-donating group) linked to an acceptor (electron-withdrawing group) via a π-conjugated system. This configuration facilitates efficient charge transfer and light harvesting.

- **Common Donors**: Triphenylamine, carbazole, and phenothiazine.
- **π-Conjugation Units**: These are often thiophene, benzothiadiazole, or fluorene derivatives which improve electron delocalization.
- **Acceptor Groups**: Typically, cyanoacrylic acid or rhodanine groups are used for anchoring onto the TiO2 surface and accepting electrons **[24].**

2.4. Porphyrin-based dyes (Metal-Free Versions)



Porphyrin derivatives are effective in sunlight harvesting due to their broad absorption spectra. Metal-free porphyrins involve tuning the structure without the central metal ion, offering comparable properties to their metal complexes **[25]**.

5. Synthesis Techniques for Metal-Free Dyes

Metal-free dyes for dye-sensitized solar cells (DSSCs) are synthesized using various techniques whereby organic compounds exhibiting desirable optical as well as electronic features are obtained. Sol-gel processes, molecular self-assembly, and traditional chemical synthesis are among the effective methods employed.

1. Sol-Gel Processes

The sol-gel process is a suitable method used in the fabrication of metal-free dyes on a laboratory scale. In simple terms, the sol-gel process is the transformation of a solution (sol) into a gel (solid-state) through the polymerization and hydrolysis of precursors. Usually, organic compounds are mixed in a solvent to obtain a sol. When the solvent starts to evaporate, there is a tendency for the sol to gelate forming a matrix suitable for the dye molecules. This technique not only assists in the morphology and the homogeneity of the dye but also improves the stability and performance of the DSSCs **[26,27].**

2.Molecular Self-Assembly

Molecular self-assembly can be termed as a more advanced technique applicable in synthesizing metal-free dyes, which relies on the forces present in molecules without the need for the application of external forces to enfold them in structured forms. This technique makes use of non-covalent interactions – hydrogen bonding, π-π stacking, van der Waals forces, etc. By careful selection of the molecular components degree of order and functionality achievable in the resulting dye structures by researchers can be high. The method is, therefore, quite appropriate for producing dyes that will have the specific light absorption characteristics and charge transfer properties required for DSSC efficiency optimization **[28].**

3. Chemical Synthesis

Chemical synthesis has remained one of the very fundamental approaches to developing metal-free dyes. This usually consists of several steps, like the formation of intermediate compounds and the final assembly of the dye structure. Condensation, cyclization, and functionalization are general organic reactions employed in the preparation of dyes with tailored properties. The possibility to modify the chemical structure enables the fine-tuning of the absorption spectrum and the electron transfer capability of the dye. Besides, during dye synthesis, the introduction of new synthetic methodologies is increasingly considered, such as principles of green chemistry, to render the process greener **[29]**.

These synthesis techniques are very important for the development of metal-free dyes and have driven research on more efficient and sustainable dye-sensitized solar cell technologies.

5. Synthesis of metal-free dyes that are used in DSSCs

1. The synthesis of 2-cyano acetanilide-derived metal-free dyes

Metal-free organic dyes have gained traction as co-sensitizers, providing cost-effective and eco-friendly solutions. Among these, 2-cyanoacetanilides have shown promise due to their distinct anchoring groups, high conjugation, and small molecular size, making them ideal candidates for co-sensitization

1.1 2-cyanoacetanilides SA7–10 as co-sensitizers in DSSCs

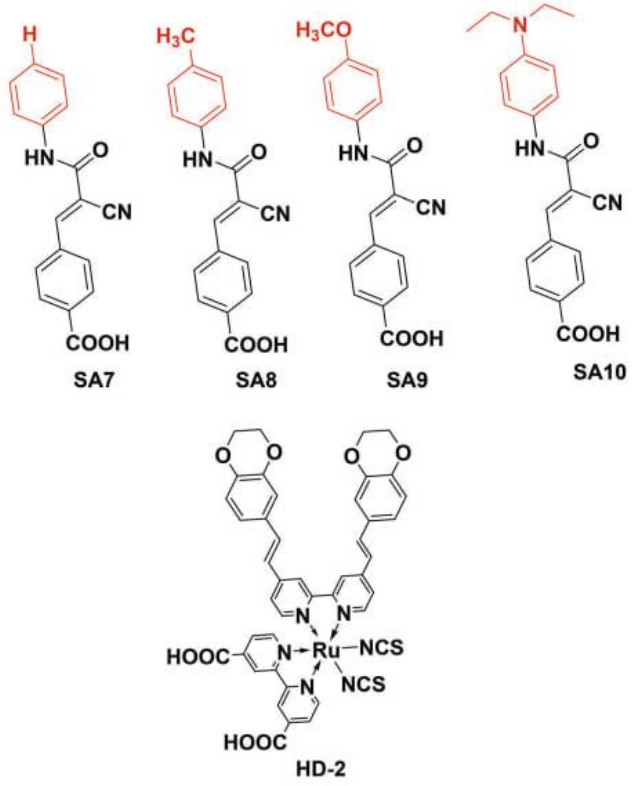


Fig. 4 Chemical structures of organic sensitizers SA7–10 and HD-2

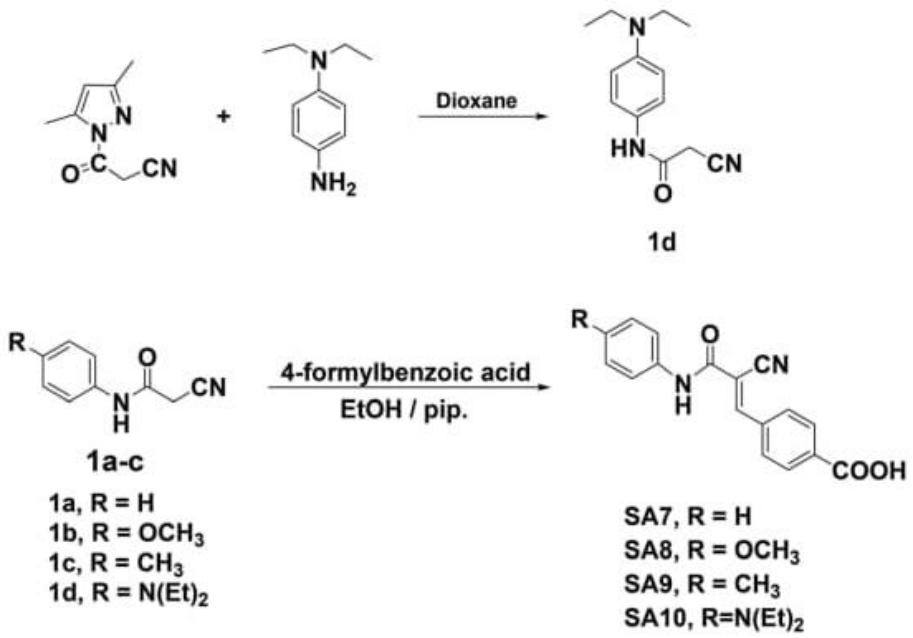
Ruthenium-based dyes like HD-2 are efficient but costly, scarce, and toxic. This drives research into low-cost, eco-friendly, and tunable metal-free organic dyes. Co-sensitization, using multiple dyes, can enhance light absorption and reduce charge recombination. This study introduces a series of 2-cyanoacetanilide-based dyes (SA7-10) as co-sensitizers for HD-2. These compact dyes with varied electron-donating groups and strong anchoring units were synthesized efficiently. Their

photovoltaic performance, photophysical properties, and electrochemical behavior in DSSCs when co-sensitized with HD-2 are investigated**.**

Synthesis of 2-Cyanoacetanilide Sensitizers (SA7–10):

The final dyes (SA7–10) were synthesized via a **Knoevenagel condensation reaction**. The reaction involved condensing **4-formylbenzoic acid** with the prepared cyanoacetanilides (1a–d) in boiling **ethanol**, using **piperidine** as a catalyst. The products were collected by filtration and recrystallized in ethanol to yield: (**SA7**: Yellow crystals) (**SA8**: Yellow crystals) (**SA9**: Orange crystals) (**SA10**: Black powder)Cyanoacetanilides (1a–d) were prepared by reacting **1-cyanoacetyl-3,5-dimethylpyrazole** with corresponding amines (e.g., aniline, 4-toluidine, 4-anisidine, and N,N-diethyl-p-phenylenediamine).

SA7–10 showed two distinct absorption bands: a π–π* transition and an internal charge transfer (ICT) band.



.**fig.5** Synthesis of cyano acetanilide sensitizers SA7–10

Among the dyes, **SA10 exhibited the smallest band gap (2.18 eV)**, the highest absorbance, and the strongest light-harvesting capability due to its strong electron-donating **N, N-diethylamino group**.

SA10 co-sensitized with HD-2 achieved the highest efficiency (8.25%), attributed to its superior light-harvesting ability and effective electron injection into the TiO_2 surface **[30]**.

1.2 2-cyanoacetanilides SA11-16 as co-sensitizers in DSSCs

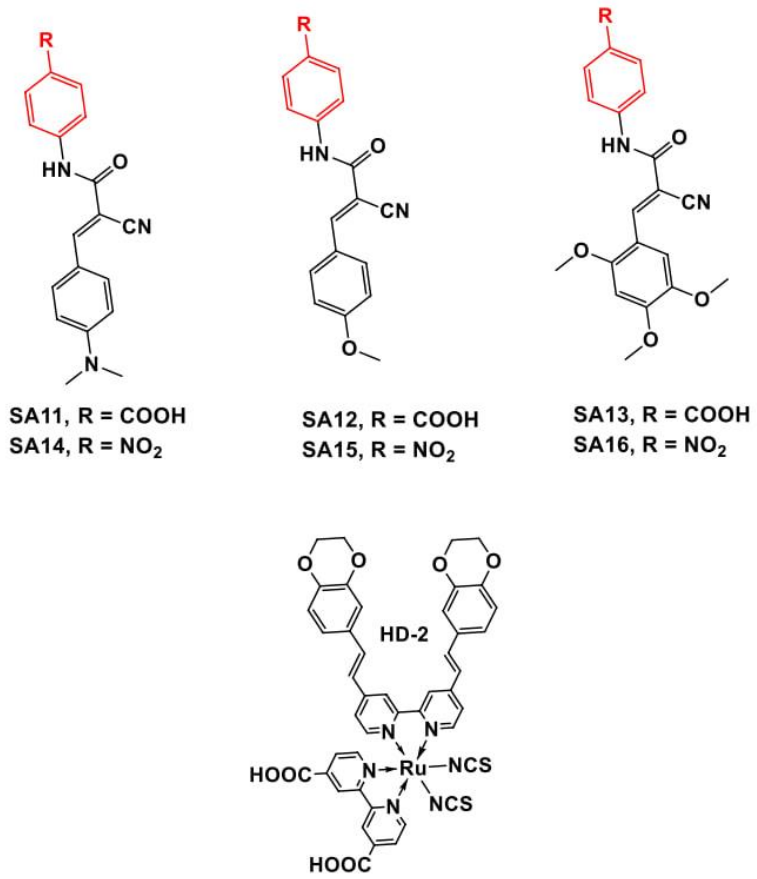


Fig.6 Chemical structures of organic sensitizers SA11-16 and HD-2

synthesis and application of six novel 2-cyanoacetanilide-derived metal-free organic dyes (SA11-SA16) as co-sensitizers for dye-sensitized solar cells (DSSCs). These dyes incorporate electron-donating groups (e.g., methoxy and dimethylamino) and anchoring groups (e.g., carboxyl (-COOH) and nitro (-NO2)). They were

synthesized through a Knoevenagel condensation reaction and characterized using spectroscopic techniques (IR, NMR, UV–Vis). Detailed theoretical studies, including DFT and TD-DFT calculations, supported the experimental findings.

Synthesis of (SA11-16):

Used Knoevenagel condensation reaction: Combined the cyanoacetanilides (1a or 1b) with different aldehydes:4-(dimethylamino) benzaldehyde - 4-methoxy benzaldehyde - 2,4,5-trimethoxybenzaldehyde

Reaction conditions:

Used ethanol as a solvent - Added piperidine as catalyst (3 drops) - Heated at 80°C under reflux for 3 hours Progress monitored by thin-layer chromatography (TLC)- Products were isolated by filtration and washed with ethanol

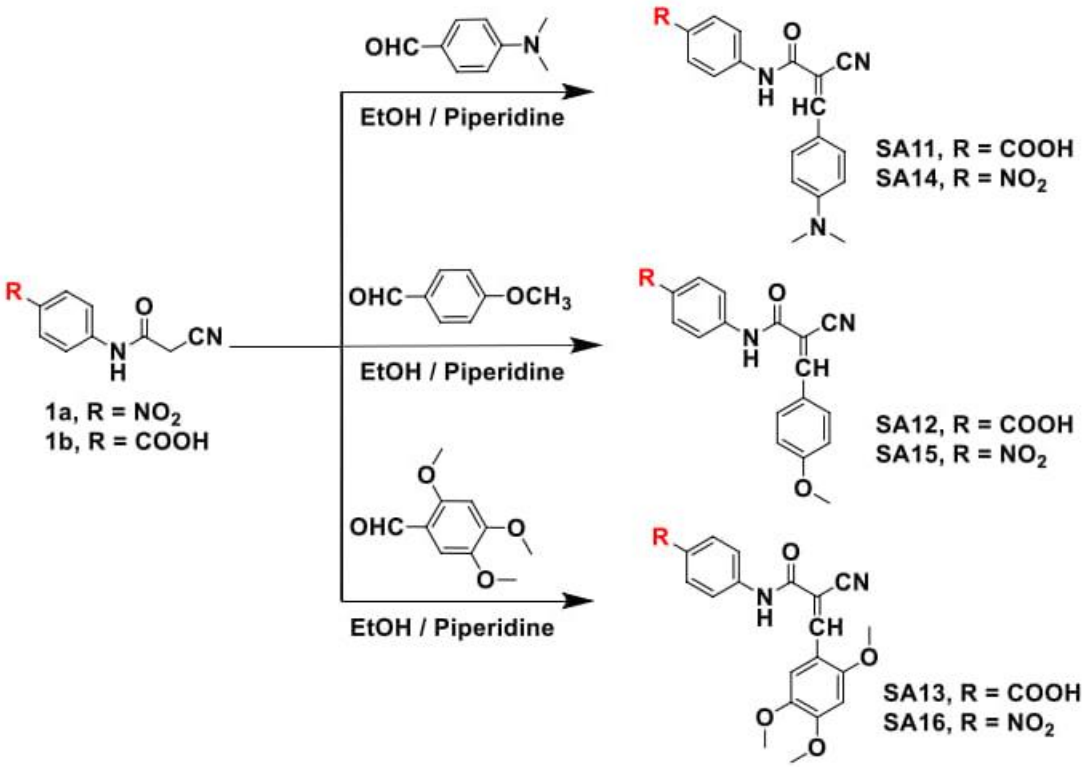


Fig7.Synthetic pathway of the SA11-16

Optical Properties: The dyes exhibited $\pi \rightarrow \pi^*$ electronic transition and intramolecular charge transfer (ICT) absorption peaks. Dyes with nitro (-NO2) as anchoring groups showed red-shifted spectra due to the strong electron-withdrawing nature of -NO2. In contrast, dyes with -COOH as anchoring groups demonstrated higher molar extinction coefficients and better optical properties for DSSCs.

Co-Sensitization with HD-2: Metal-free dyes were combined with HD-2, a ruthenium-based dye with high photovoltaic performance but limited light absorption. Co-sensitization improved DSSC performance, with SA12 achieving an 8.69% power conversion efficiency (PCE) compared to HD-2 alone (7.46%). This enhancement is attributed to better surface coverage, reduced dye aggregation, and improved light absorption.

Anchoring Group Performance: Dyes with -COOH groups (e.g., SA12) showed superior photovoltaic performance compared to those with -NO2 groups (e.g., SA14), likely due to better anchoring on the TiO2 surface and more efficient electron injection **[31].**

1.3.(N-benzothiazolyl)-cyanoacetamides SA17–20as co-sensitizers in DSSCs

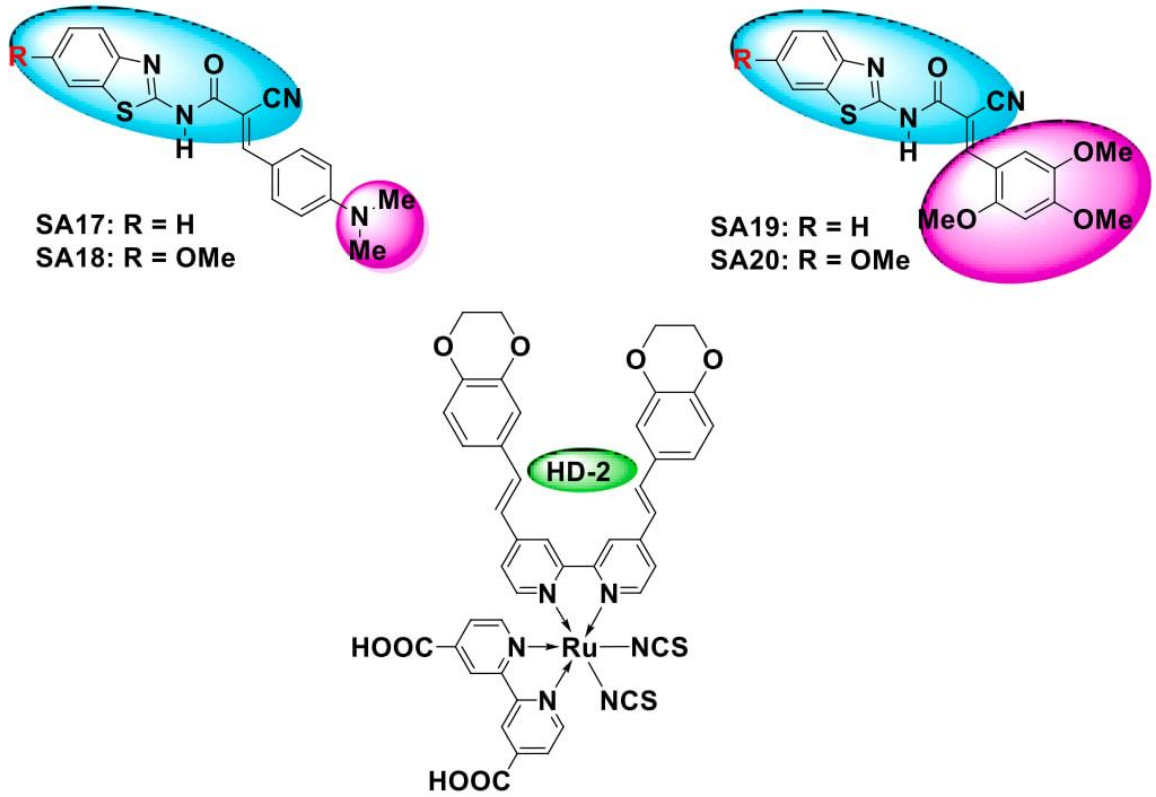


Fig.8 Molecular structures of SA17–20 and HD-2

Four novels (N-benzothiazole)-cyanoacetamide-based dyes (SA17-20) were synthesized as co-sensitizers for HD-2 to investigate the impact of their structural features on DSSC performance. [1] These dyes, incorporating strong donors and anchoring groups, were synthesized via Knoevenagel condensation. [2] Their ability to reduce dye aggregation, suppress charge recombination, and enhance DSSC efficiency was evaluated. SA20 demonstrated the highest efficiency at 8.27%, highlighting the potential of (N-benzothiazole)-cyanoacetamides as cost-effective and high-performance co-sensitizers for next-generation DSSCs

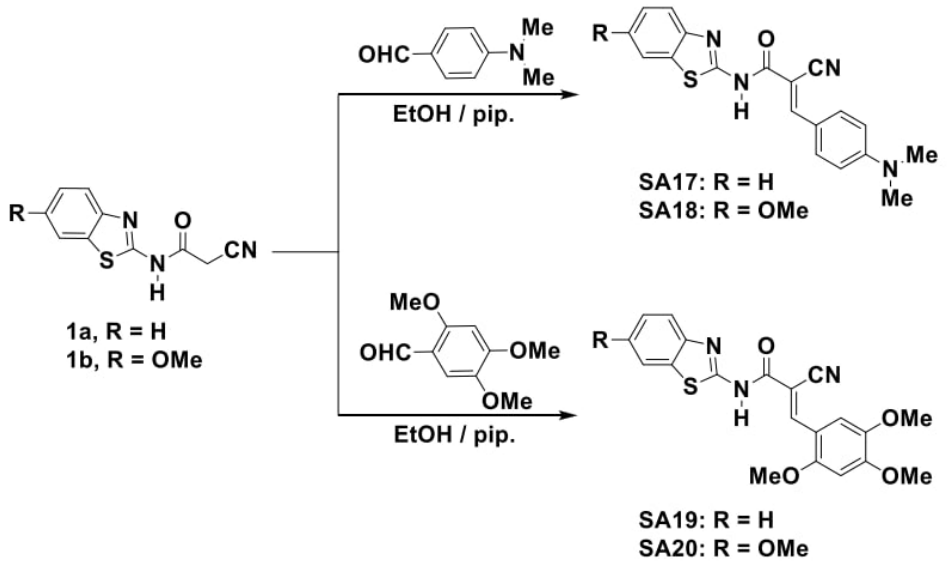


Fig9. Synthetic route of (N-benzothiazolyl)-cyanoacetamide co-sensitizers SA17–20

Preparation of Starting Materials:

(N-Benzothiazolyl)-cyanoacetamides (1a and 1b) were synthesized by refluxing 2-aminobenzothiazole or 2-amino-6-methoxybenzothiazole with 1-cyanoacetyl-3,5-dimethylpyrazole in dioxane.

Knoevenagel Condensation Reaction:

The target dyes (SA17-20) were obtained through a Knoevenagel condensation reaction between (N-benzothiazole)-cyanoacetamides 1a/b and aldehyde derivatives (e.g., 4-(dimethylamino) benzaldehyde, 2,4,5-trimethoxybenzaldehyde) in ethanol using piperidine as a catalyst.

Characterization of SA20:

SA20 exhibited a red-shifted absorption spectrum and the highest molar extinction coefficient, attributed to the presence of strong donating groups (trimethoxybenzene).

Synthetic Efficiency:

High yields (78-86%) were achieved for all dyes, with SA20 demonstrating the most promising photophysical and photovoltaic properties **[31]**.

2. The synthesis of carbazole additives in metal-free dyes

Carbazole dyes are promising DSSC sensitizers due to their electron-donating properties, stability, and efficient charge transfer. Synthesis often involves Suzuki or Stille coupling to incorporate carbazole into organic dye structures. These dyes enhance light absorption and DSSC performance, offering a sustainable and cost-effective alternative to metal-based dyes.

2.1carbazole MES 1-4 as co-sensitizers in DSSCs

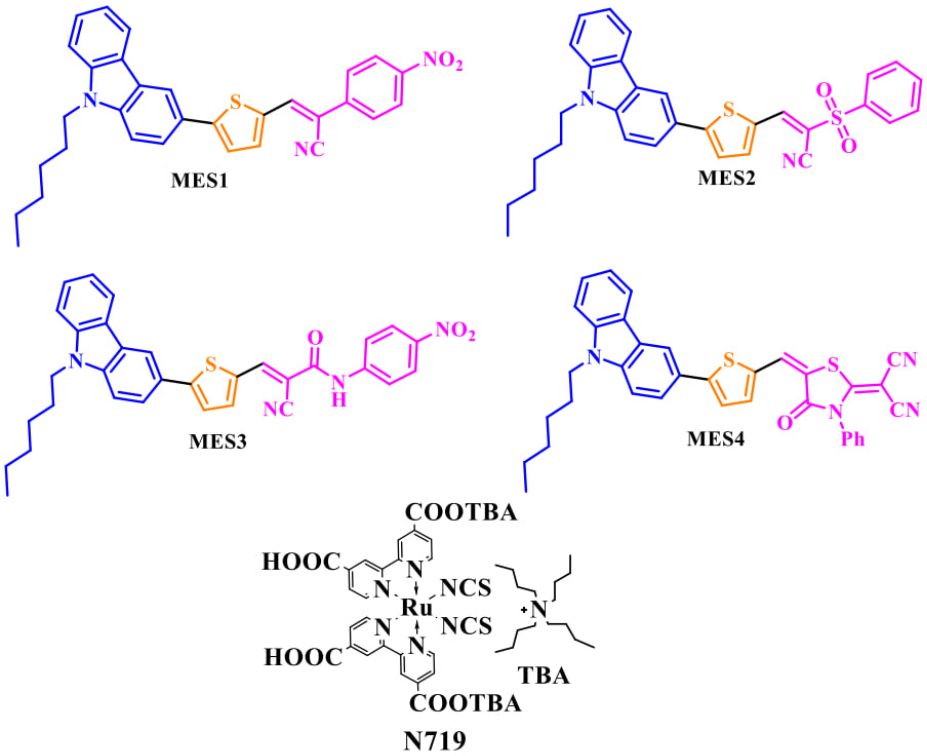


Fig. 10. Chemical structures of sensitizers MES1-4.

Carbazole-based D-π-A dyes (MES1-MES4) with varied acceptors (nitrophenyl acetonitrile, sulphonyl acetonitrile, thiazolidine-4-one) were synthesized for DSSCs using bromination, Suzuki coupling, and Knoevenagel condensation.

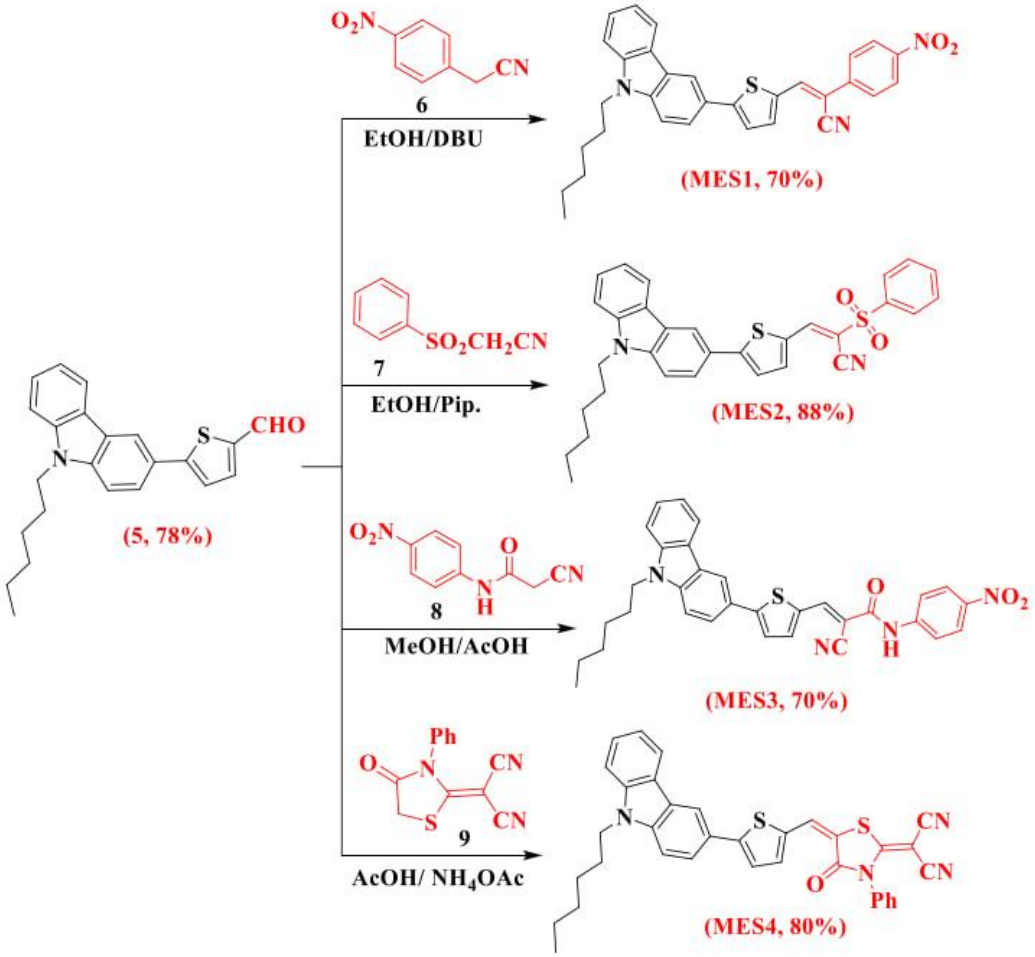


Fig.11. Synthetic pathway of carbazole-based sensitizers MES1-4.

Preparation of Carbazole Derivatives: The starting material, 9-hexylcarbazole, was brominated to produce 3-bromo-9-hexyl-9H-carbazole.

A Suzuki coupling reaction was performed to link the carbazole derivative to a thiophene-2-carbaldehyde unit, forming the intermediate compound 5-(9-hexyl-9H-carbazol-3-yl) thiophene-2-carbaldehyde.

Final Dye Formation via Knoevenagel Condensation: The intermediate was reacted with various electron-withdrawing acceptor/anchoring groups (e.g., nitrophenylacetonitrile, sulphonylacetonitrile, and thiazolidine-4-one) to yield the final dyes MES1-MES4 **[33]**.

3.carbazole-based cyanoacetanilides (5a-c) and carbazole based thiazolidine derivatives (7a and 7b)

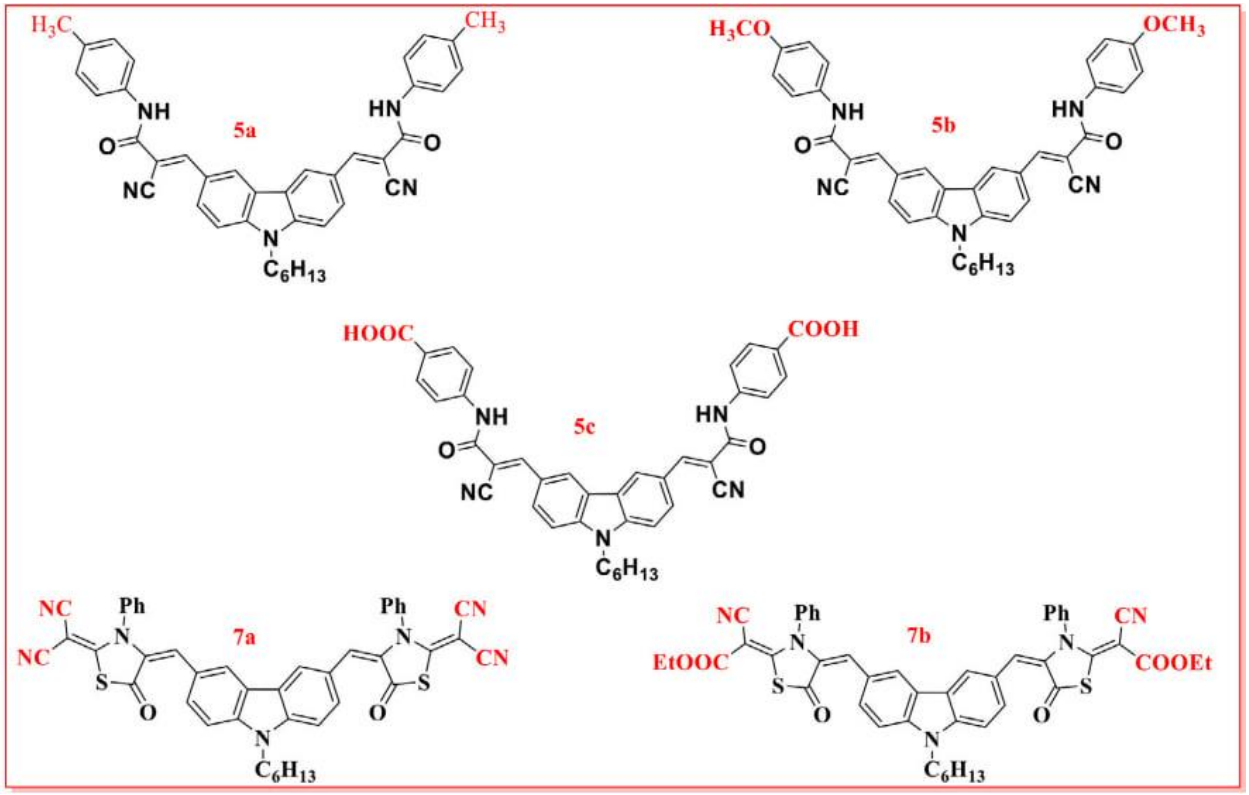


Fig.12. Chemical structure of design a new class of carbazole-based compounds (5a-c and 7a-b)

Novel carbazole-based dyes were synthesized through a multi-step process. Starting with 9H-carbazole, N-hexyl carbazole was prepared, followed by formylation to yield 3,6-diformyl-N-hexylcarbazole. This intermediate underwent Knoevenagel condensation with cyanoacetamide derivatives to produce bis(cyano-acrylamide) dyes, and with thiazolidine-5-one derivatives to yield bis(phenylthiazolidinyl) dyes. Spectroscopic techniques confirmed the structures of the synthesized dyes. These compounds were designed with functionalities to enhance their performance in photovoltaic devices and explore their potential as anticancer agents.

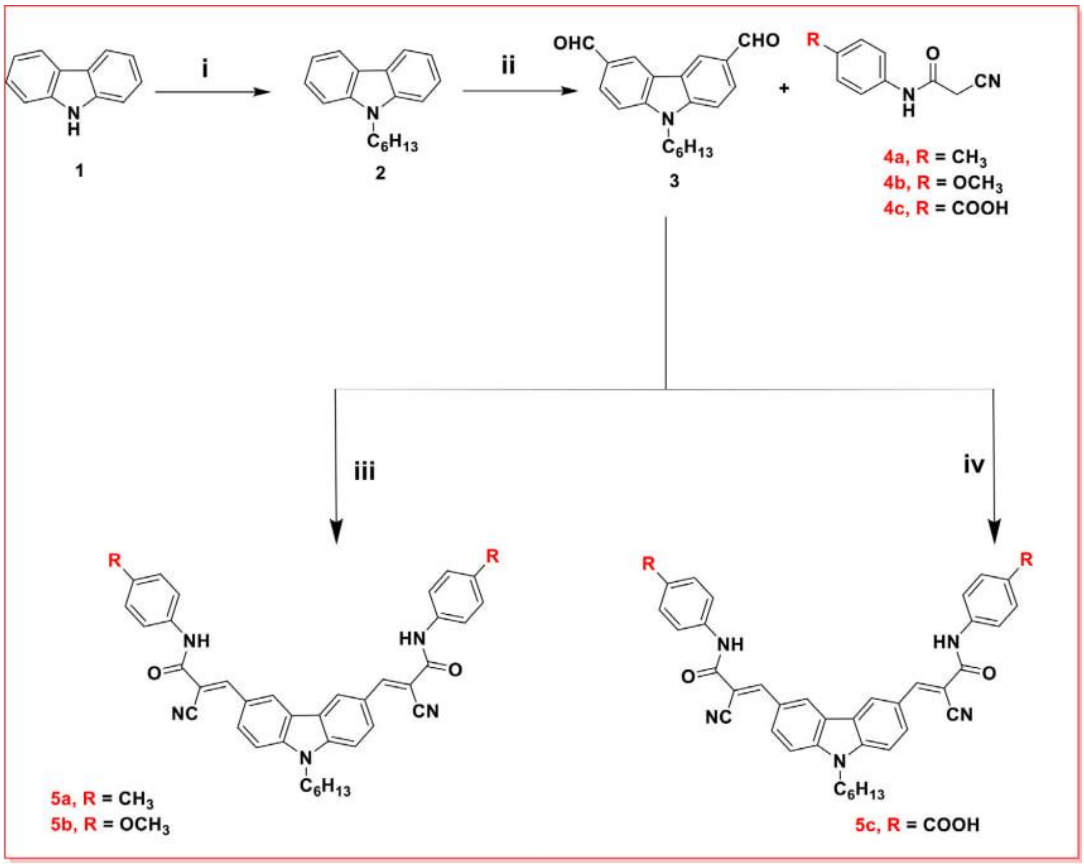


Fig13. Synthetic routes for bis(cyano-acrylamide)carbazole 5a-c: (i) 1-bromohexane, NaOH, DMSO, rt (ii) POCl3, DMF (iii) Ethanol, pip (iv) acetic acid

A new class of carbazole-based dyes (5a-c and 7a-b) was synthesized and evaluated for their potential as DSSC sensitizers. Notably, compound 7b, which incorporates a thiazolidine-5-one moiety, exhibited superior photovoltaic performance compared to other dyes. Theoretical and experimental investigations revealed that 7b possesses a wide absorption range, a low energy gap (2.31 eV), and a strong electron injection capability, making it the most favorable candidate for DSSC applications. This introduction aims to explore the structural features, optical properties, and electronic behavior of the highest-performing dye, 7b, and its potential impact on enhancing DSSC efficiency**[34].**

4. The synthesis of triphenylamine-derived metal-free organic dyes

Triphenylamine (TPA)--based dyes are promising DSSC sensitizers due to their strong electron-donating properties, enhancing charge injection and light absorption. The TPA core facilitates electron transfer to TiO_2, while the conjugated structure broadens light absorption. Modifying the TPA structure with electron-withdrawing/donating groups (cyano, thiophene, carbazole) fine-tune their properties. These dyes offer a sustainable and cost-effective alternative to ruthenium-based sensitizers

4.1Triphenylamine SFA-5–8as co-sensitizers in DSSCs

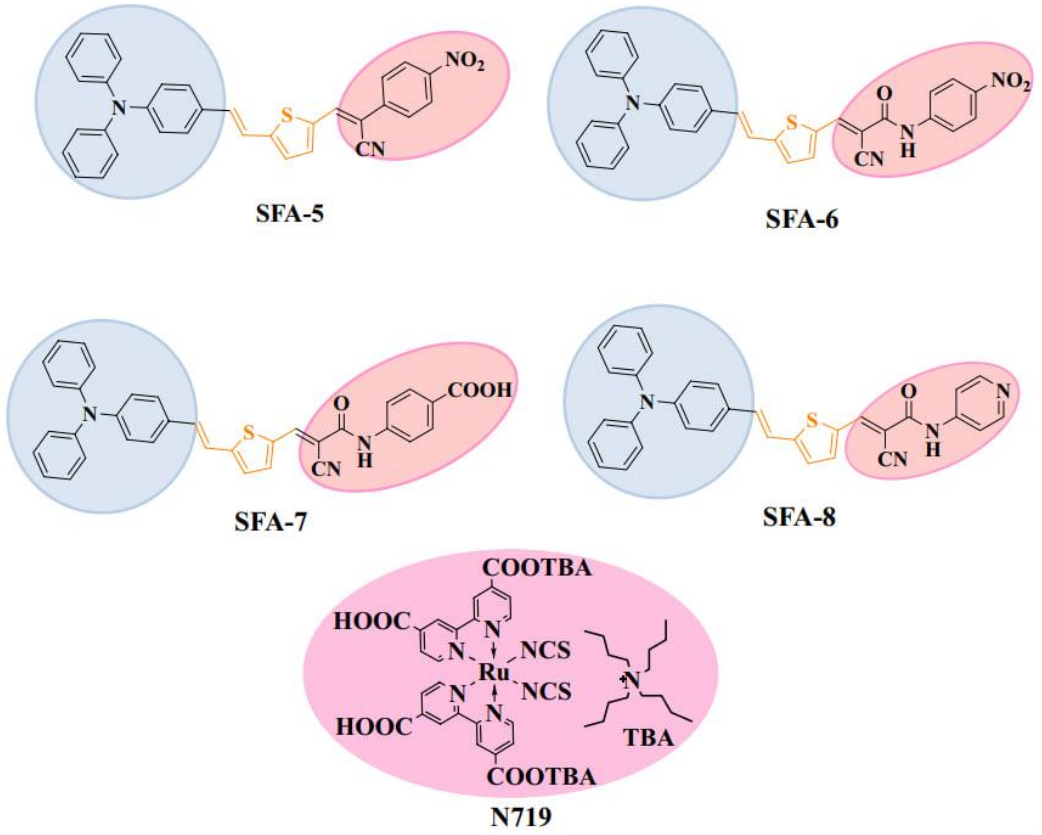


Fig14. Molecular structures of sensitizers SFA-5–8 and N719

Triphenylamine-based D-π-A dyes (SFA-5 to SFA-8) were synthesized. Thiophene acts as the π-bridge, while acceptors include phenyl acetonitrile and cyanoacetamide derivatives with electron-withdrawing groups (COOH, NO2, CN, pyridyl). The

synthesis involved preparing 5(4(diphenylamino)styryl) thiophene-2-carbaldehyde, followed by Vilsmeier-Haack reaction and Knoevenagel condensation with methylene compounds.

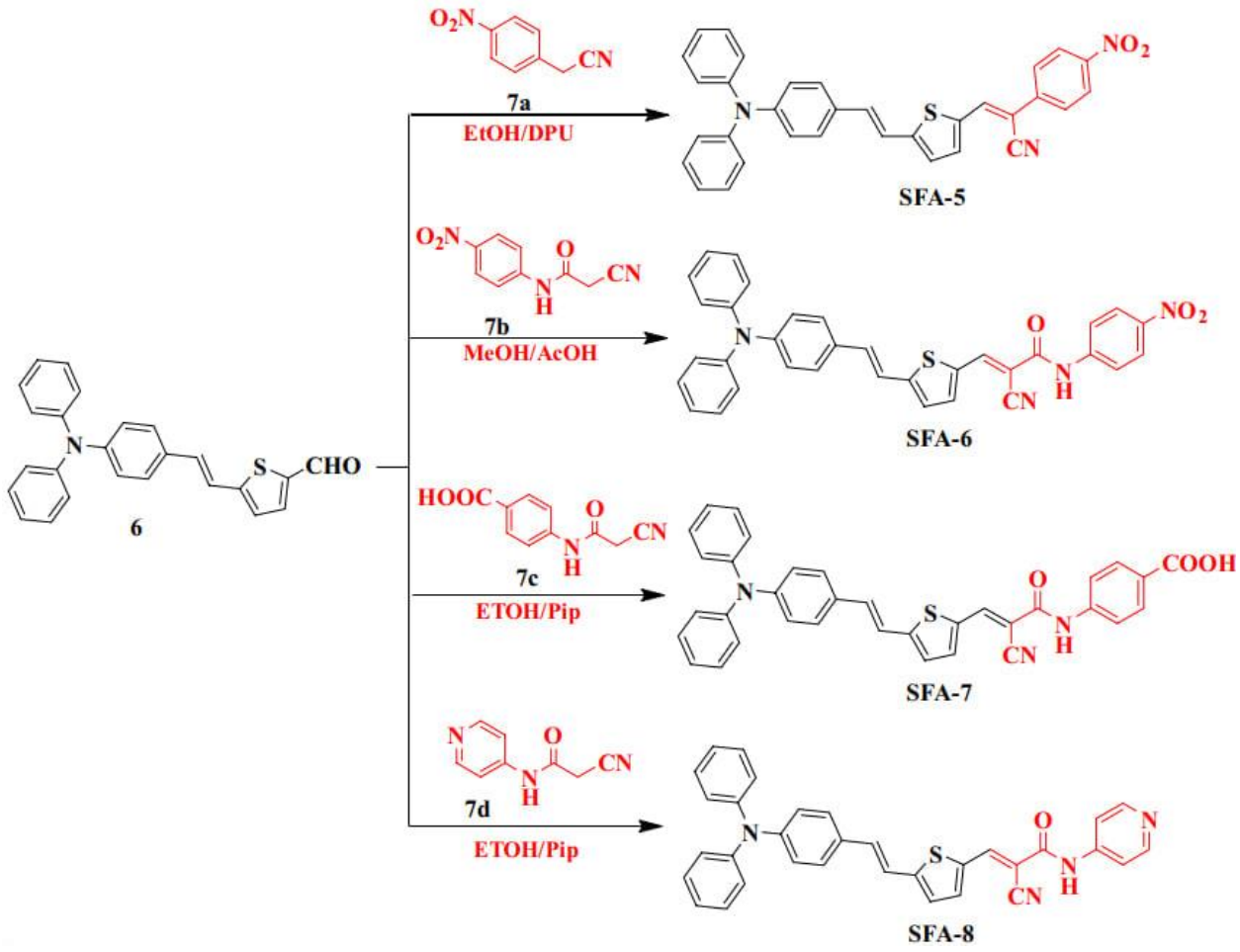


Fig15. Synthetic routes of SFA-5–8 sensitizers

PA-based dyes are characterized by their excellent electron-donating properties, structural flexibility, and resistance to aggregation. In this study, we report the design and synthesis of four novel triphenylamine-based dyes (SFA-5 to SFA-8) featuring a donor–π-bridge–acceptor (D–π–A) architecture. These dyes incorporate thiophene as a conjugated linker and phenylacetonitrile or cyanoacetamide derivatives as electron-accepting groups. The inclusion of multiple anchoring groups (COOH, CN, NO2, and pyridyl) aims to enhance dye adsorption on the TiO2 surface, improve electron injection efficiency, and suppress charge recombination **[35]**.

5. (TMS-1-4) based on triphenylamine (TPA) and carbazole (Cz)

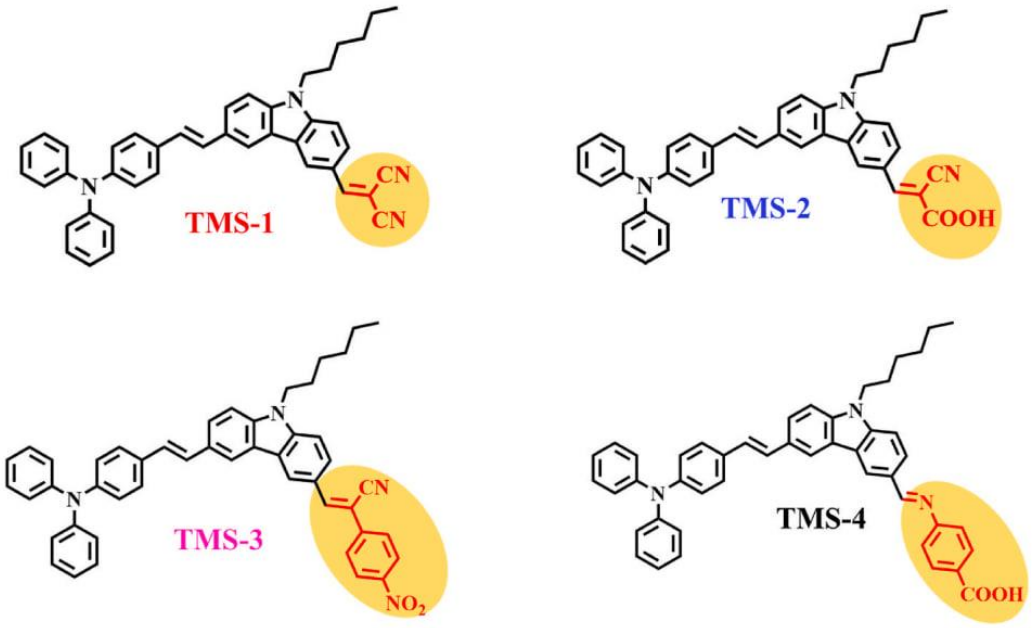


Fig.16. Chemical structures of sensitizers TMS-1-4

Preparation of a series of novel metal-free organic dyes (TMS-1 to TMS-4) with a D-D-π-A structure, designed for dye-sensitized solar cells (DSSCs). The process begins with the synthesis of a key intermediate, 7-(4-(diphenylamine) styryl)-9-hexyl-9H-carbazole-3-carbaldehyde (compound 6), through a Wittig reaction followed by a Vilsmeier-Haack formylation. This intermediate is then subjected to Knoevenagel condensation with various electron-accepting groups, including malononitrile (-MN), cyanoacrylic acid (-CA), nitrophenylacetonitrile (-NAN), and iminophenylcarboxylic acid (-CH), to produce the final dyes TMS-1, TMS-2, TMS-3, and TMS-4, respectively. The dyes were purified and characterized using spectroscopic techniques such as UV-Vis, NMR, and IR, confirming their structures and properties. These dyes were specifically designed to enhance light absorption, electron injection, and photovoltaic performance in DSSCs.

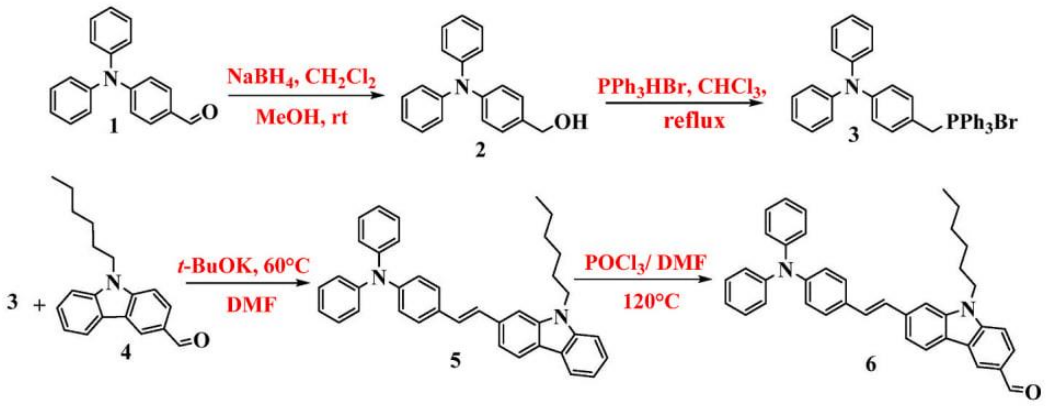


fig17. Synthetic procedures of triphenylamine-carbazole aldehyde 6.

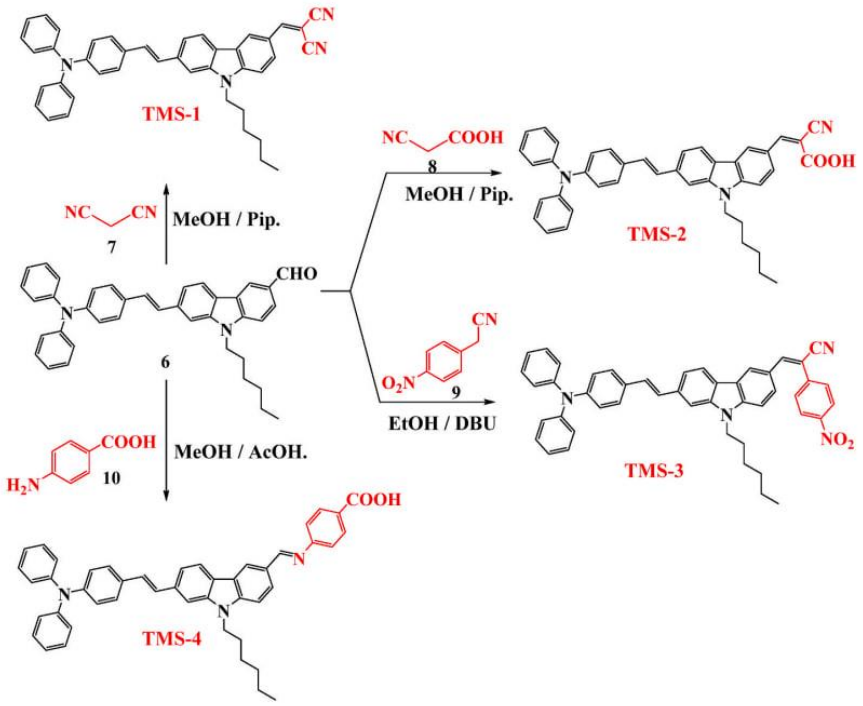


Fig.18.Synthetic route of sensitizers TMS-1-4.

A series of novel metal-free organic dyes (TMS-1 to TMS-4) with a D-D-π-A structure were synthesized, incorporating triphenylamine and carbazole as strong electron-donating units and various electron-accepting groups. Among these dyes, **TMS-2**, which features a cyanoacrylic acid (-CA) acceptor, demonstrated the highest photovoltaic performance with a power conversion efficiency (PCE) of 8.81%. This superior performance is attributed to its strong light-harvesting ability, efficient electron injection, and reduced recombination losses. The synthesis,

characterization, and application of these dyes provide valuable insights into the development of high-performance sensitizers for DSSCs **[36].**

6. Conclusion:

In conclusion, dye-sensitized solar cells (DSSCs) represent a promising frontier in renewable energy technologies due to their unique advantages, including cost-effectiveness, transparency, and the ability to function efficiently under low-light conditions. By mimicking natural photosynthesis, DSSCs have opened new avenues for harnessing solar energy with innovative materials and designs. While there are challenges related to their scalability, stability, and efficiency, ongoing research into novel dyes, photoanodes, electrolytes, and counter electrodes continues to address these limitations. The development of sustainable and high-performance sensitizers, including metal-free organic dyes and natural dyes, is particularly noteworthy, offering environmentally friendly alternatives with enhanced photovoltaic properties. As the field advances, DSSCs are poised to play a significant role in the global transition towards clean and renewable energy sources, ensuring a sustainable future for generations to come.

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